DEPARTMENT OF CSE · TKM COLLEGE OF ENGINEERING · FINAL EVALUATION

AURA.

Anonymous campus issue reporting that students trust and administrators can act on.

REPORT · TRACK · RESOLVE · IMPROVE

TEAM LEAD
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HYPE ENGINE
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SECURITY & VAULT
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DATABASE & GUI
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RECAP · WHY AURA EXISTS

Campus problems get reported. Then they disappear.

TODAY

No tracking

WhatsApp groups, a paper box and flat Google Forms. Students never learn if anything happened.

TODAY

No priority

The loudest complaint wins, not the one affecting the most students.

TODAY

Fear of reprisal

A named complaint about a lab, hostel or office can follow a student. So it goes unreported.

AURA

5-state lifecycle

PENDING → ASSIGNED → IN_PROGRESS → RESOLVED / REJECTED, with a public resolution note and an audit trail.

AURA

Hype engine

One upvote per verified student pushes the most widespread problems to the top of Trending.

AURA

Anonymity vault

Reports carry no author. Only the student who filed it can see their own receipt.

SYSTEM INTEGRATION P03

Two front doors. One source of truth.

A report filed on a phone shows up in the admin's desktop triage table. A status change on the desktop shows up on the student's phone.

DESKTOP · JAVA 17

FlatLaf Swing app

UI → Service → DAO → JDBC. Prepared statements only. BCrypt sign-in.

WEB + PHONE · PWA

Vercel portal

Supabase Auth sign-in. Works offline-installable on any phone browser.

JDBC
via pooler

6 RPCs
role-checked

SUPABASE · POSTGRESQL

6 tables, constraints enforced in the database

users	submissions	student_submission_receipts
submission_hype	resolution_notes	submission_history

Browser roles cannot touch tables directly; they can only call the 6 portal functions, each of which re-checks identity, domain and role.

39 Java classes · 3,676 lines · 548 lines SQL

docs/ARCHITECTURE.md · sql/web_portal.sql

MODULES · ALL INTEGRATED, ALL TESTED P03

Five modules, one pipeline

01 · NAFIH
Auth & Vault
`AuthService`, `ValidationUtil`, `PasswordUtil`, `StudentReceiptDAO`

tested

→

02 · RAHANDEEP
Schema, DAO & GUI
`schema.sql`, `SubmissionDAO`, student UI panels

tested

→

03 · NIRMAL
Hype & Trending
`HypeService`, `SubmissionHypeDAO`, one-vote constraint

tested

→

04 · ATHIL
Lifecycle & Reports
`TrackingService`, `ReportService`, history, CSV

tested

→

05 · RINSHID
Cloud, Web & CI
`DatabaseConfig`, web portal, RPC security, CI/CD

tested

Student signs in → files a report → gets a private receipt → upvotes others.

Admin sees an author-free queue → assigns → resolves with a note → exports CSV.

Every state change lands in `submission_history` with old state, new state and who changed it.

Module ownership agreed in Phase 1

README.md §3

THE DECOUPLED ANONYMITY VAULT P03 P07

Even the administrator cannot find out who reported it.

STUDENT SIDE

My Receipts

Student sees only rows where `student_id = me`. Other students' receipts are invisible.

SUBMISSIONS TABLE

Title · category · location · status · hype

No author column. Not hidden, not encrypted: it does not exist.

ADMIN SIDE

Triage queue

Admin code never queries receipts. The receipt function rejects an ADMIN session.

Reflection test

Build fails if anyone adds a student/user field to `Submission`.

Database test

An admin calling `portal_list_my_submissions()` gets a permission error.

Export test

CSV header contains no student, email, user or receipt column.

AdminAnonymityRegressionTest · db-portal.test.js · ReportAndTrackingServiceTest

docs/SECURITY.md

LIVE DEMONSTRATION P03

One report, end to end, across both apps

1 · PHONE
Sign in
Non-`@tkmce.ac.in` email rejected

→

2 · PHONE
Files a report
Private receipt appears in My Receipts

→

3 · PHONE
Others upvote
Second vote toggles, never duplicates

→

4 · DESKTOP
Admin triage
Same report, no author shown

→

5 · DESKTOP
Resolve
Assign → resolve; note and history logged

→

6 · PHONE
Student sees it
RESOLVED + note; admin exports CSV

Fallback: mobile hotspot if campus Wi-Fi fails; if the database is unreachable, a screen recording of this run made the day before, announced as a recording.

MR

INDIVIDUAL CONTRIBUTION P03 P08

Muhammed Rinshid VP · Team Lead

I BUILT

— Layered architecture and `DatabaseConfig` for Supabase

— Live web portal with Supabase Auth

— `sql/web_portal.sql`: 6 role-checked RPCs

— GitHub Actions CI/CD → Vercel

CONCEPT I CAN DEFEND

Least privilege for the browser

The browser holds only the public anon key. Direct table access is revoked; each RPC re-checks identity, `@tkmce.ac.in` domain and role. The DB password never leaves the server.

PROOF IT WORKS

— `db-portal.test.js`: student, second student, admin, outsider

— `config-api.test.js`: only public URL + key exposed

— Pipeline runs both suites on every push

Leadership: module split, integration of all five modules, presentation lead

.github/workflows/ci-cd.yml

NB

INDIVIDUAL CONTRIBUTION P03 P08

Nirmal Binoy · Hype & Trending Engine

I BUILT

— `HypeService` and `SubmissionHypeDAO`

— Trending ranking by verified upvotes

— UML class diagram and domain models

CONCEPT I CAN DEFEND

Rules that live in the database

A Java "already voted?" check can race. `UNIQUE (submission_id, student_id)` makes a duplicate vote impossible even when two clicks arrive at once.

PROOF IT WORKS

— `SubmissionAndHypeServiceTest`: duplicate vote rejected

— Web test: second click un-hypes instead of double-counting

Constraint: **uq_hype_once_per_student** in sql/schema.sql

Phase 1 UML · docs/ARCHITECTURE.md

MN

INDIVIDUAL CONTRIBUTION P03 P08

Mohammed Nafih · Security & Anonymity Vault

I BUILT

— `AuthService`, `ValidationUtil`, `PasswordUtil`

— `StudentReceiptDAO`: the vault boundary

— First DB submission workflow; technical learning manual

CONCEPT I CAN DEFEND

Fail closed, leak nothing

Anchored domain regex. BCrypt, work factor 10, fresh salt. Unknown email and wrong password give the **same** error, so the login form can't reveal who is registered.

PROOF IT WORKS

— `ValidationAndPasswordUtilTest`: look-alike domains rejected

— `AuthServiceTest`: errors indistinguishable

— Anonymity regression tests (slide 5)

Design change: Google OAuth retired; a desktop JAR can't keep a client secret

docs/SECURITY.md

RD

INDIVIDUAL CONTRIBUTION P03 P08

Rahandeep RD · Database & Desktop GUI

I BUILT

— `schema.sql`: 6 tables, FKs, CHECKs, indexes

— `SubmissionDAO` and the DAO pattern

— FlatLaf student dashboard and report form

CONCEPT I CAN DEFEND

Atomic, injection-proof writes

Submission + receipt are inserted in **one transaction**: both commit or neither does. Every query is a `PreparedStatement`; no SQL is ever built from user text.

PROOF IT WORKS

— `SubmissionAndHypeServiceTest`: pipeline round-trip

— `StatusColorContrastTest`: every status badge ≥ 4.5:1 (WCAG AA)

Packages: **aura.dao.*** · **aura.ui.student.***

docs/DATABASE.md

AR

INDIVIDUAL CONTRIBUTION P03 P08

Athil Rahuman A · Lifecycle, Reports & Testing

I BUILT

— `TrackingService`: 5-state transitions + audit

— `ReportService`: KPI counts, CSV export

— First Supabase connection; progress tracking

CONCEPT I CAN DEFEND

An audit trail you can trust

Each real change logs old → new state and who did it. No-op changes log nothing; unknown tickets are rejected. CSV follows RFC 4180 quoting.

PROOF IT WORKS

— `ReportAndTrackingServiceTest`: 5 tests

— SQL test: PENDING → RESOLVED history row

Packages: **aura.service.TrackingService** · **ReportService**

docs/TESTING.md

VERIFICATION P03 P05

41 automated tests, run on every push

16

Java unit tests

Run offline: domain gate, BCrypt, CSV, audit trail, badge contrast, no user enumeration

8

Java DB tests

Against live Supabase: pipeline, single vote, vault

4

SQL / API tests

Real migration on in-memory Postgres; config endpoint

13

Browser tests

Playwright, desktop + phone: sign-in gate, student, admin, expiry

Pipeline: push → compile Java 17 → JUnit 5 → Node + Playwright → build fat JAR → deploy to Vercel. 0 failures live sign-in: human checklist

Result on 18 Oct 2026: Java 24 / 24 pass on the live DB · Web 17 / 17 pass

docs/TESTING.md

TOOLS ACROSS EVERY PHASE P05

A modern tool for every phase of the work

PHASE	TOOLS	WHY THIS ONE
Requirements	Markdown PRD / TRD in Git, UN SDG mapping	Versioned with the code; reviewable in pull requests
Design	UML use-case, class, sequence diagrams · Mermaid Gantt	Phase 1 approved model drives the Java packages
Build: desktop	Java 17 · Maven · FlatLaf · JDBC (pgJDBC) · jBCrypt	Syllabus-aligned, one-command build
Build: web + data	Supabase Postgres + Auth · supabase-js · PWA · Vercel	Managed DB and auth; free tier; phone-installable
Testing	JUnit 5 · Node test runner · PGlite · Playwright	Real SQL and real browser, not mocks of our own code
DevOps	Git · GitHub · GitHub Actions · Vercel deploy	Every push tested and deployed automatically
Assistance	AI coding assistant governed by `AGENT.md`	Same tests and review gate as human-written code

IMPACT OF ENGINEERING ON SOCIETY P06

Who it helps, what could go wrong, how we'd know

BENEFIT

Voice without risk

— Students report safety, hygiene and lab issues without fear

— Admins get a ranked queue, not a WhatsApp scroll

— Public resolution notes build institutional trust (SDG 16)

— Fewer lost lab hours from broken equipment (SDG 4)

RISK & MITIGATION

Anonymity can be abused

— Spam / false reports → verified `@tkmce.ac.in` accounts only, one vote each

— Pile-ons → admin can REJECT with a public reason

— Unequal access → works on any phone browser, no app install

HOW TO EVALUATE

Metrics already recorded

— Median time PENDING → RESOLVED (from `submission_history`)

— % resolved vs rejected per category

— Repeat reports on the same location

not yet measured: pre-rollout

UN SDG 16 Strong institutions · 11 Safe communities · 9 Infrastructure · 4 Quality education

docs/PRD.md §2

ETHICS AND INCLUSIVITY P07

Ethical issues we hit, and what we did

Privacy vs accountability

Chose structural anonymity (no author column) over "trust the admin". Abuse is checked by verified accounts and one vote per student, not by identifying people.

Collect only what's needed

Passwords stored only as BCrypt hashes. CSV export carries no identity (tested). Matches the purpose-limitation principle of India's DPDP Act 2023.

Never mislead users

No sample or fake data: every screen shows the live database. The UI never shows success unless the database confirmed it.

Inclusive by default

WCAG AA contrast, light + dark themes, phone-first layout, free to use. **Known gap:** staff without a `@tkmce.ac.in` email can't report yet.

TEAM STRUCTURE AND COLLABORATION P08 P010

Clear ownership, one shared contract

MEMBER	ROLE	OWNS END-TO-END	PRESENTS TODAY
Muhammed Rinshid VP	Team Lead	Architecture, cloud DB, web portal, CI/CD	Architecture, demo, cost
Nirmal Binoy	Core Developer	Hype engine, trending, UML model	Society impact, future work
Mohammed Nafih	Core Developer	Auth, BCrypt, domain gate, anonymity vault	Anonymity vault, ethics
Rahandeep RD	Database & GUI	Schema, DAO layer, FlatLaf student UI	Tools, self-learning
Athil Rahuman A	Docs & Process	Lifecycle, reports, CSV, testing	Testing, team, timeline

Shared contract: one schema (`sql/schema.sql`) used by both the desktop and web apps.

Git workflow: work merged into `main`; CI runs both test suites on every push and PR.

Definition of done: compiles clean, has a JUnit test, matches the UML, keeps the vault (`AGENT.md` §6).

Multidisciplinary: database · security · UI/UX · DevOps · documentation

README.md §3

PLAN VS ACTUAL P010

10-week plan, followed, plus one extension

Dates are from the Git history and the extension's task log.

25 AUG
Repo + requirements
PRD, TRD, scope approved

05 SEP
Architecture + schema
UML, 6-table Supabase schema

06–11 SEP
DAO, services, GUI
In-memory → live DB; CI pipeline

13 SEP
Phase 2 review
Core system demonstrated

24 SEP – 12 OCT
Extension: live portal
Supabase Auth, RPC security, Playwright

17 OCT
Final evaluation
Integrated, tested, deployed

10 / 10
planned weekly milestones delivered

+1 extension
live web portal with real authentication, beyond the original plan

1 design change
Google OAuth → email/password, for security (slide 9)

Gantt chart and planned-vs-completed matrix

docs/ROADMAP.md

COST SPECIFICATION P010

₹0 to build. About ₹4,300 a month to run.

DEVELOPMENT (ACTUAL)

Supabase (DB + Auth)	Free tier	₹0
Vercel hosting	Hobby tier	₹0
GitHub + Actions CI	Public repo	₹0
JDK, Maven, FlatLaf, JUnit, Playwright	Open source	₹0
Team effort	5 × 10 wk × 8 h	~400 h
Notional labour value	**at ₹200 / h**	**≈ ₹80,000**

CAMPUS DEPLOYMENT (ESTIMATE)

Supabase Pro	Daily backups, no auto-pause	~US$25 / mo
Vercel Pro	Commercial use, 1 seat	~US$20 / mo
Domain (optional)	e.g. aura.tkmce.ac.in	₹0 (college DNS)
Total		**≈ ₹4,300 / mo**

Alternative: self-host PostgreSQL on a college server. Licence cost ₹0; only hardware and admin time.

List prices, Oct 2026 · US$1 ≈ ₹95

Labour rate is notional, for planning only

ADAPTABILITY AND SELF-LEARNING P011

What we taught ourselves, and when we changed course

LEARNED BEYOND THE SYLLABUS

- Cloud PostgreSQL over JDBC, connection management
- Postgres grants and `SECURITY DEFINER` functions
- OAuth 2.0 PKCE (RFC 8252) and JWT sessions
- Browser end-to-end testing with Playwright
- CI/CD with GitHub Actions; PWAs and service workers

TIMES WE CHANGED COURSE

- **In-memory → live DB** (6 → 11 Sep): the prototype service was replaced by real DAOs once the schema stabilised
- **Google OAuth → email/password:** a desktop JAR cannot keep an OAuth client secret secret
- **Local-storage web demo → real RPCs:** the portal must never show success that the DB did not confirm

LIMITATIONS AND FUTURE WORK

What it doesn't do yet

LIMITATION

Account provisioning

No self-sign-up by design; the college must create accounts in Supabase Auth before live use.

LIMITATION

No notifications

Students must open the app to see a status change.

LIMITATION

Email-only access

Contract staff without a college email can't report.

NEXT

Push notifications

Service worker is already installed; add Web Push on status change.

NEXT

Photo evidence

`photo_url` column exists; add Supabase Storage upload.

NEXT

Resolution analytics

Time-to-resolve dashboards from `submission_history`.

CONCLUSION

AURA is working, integrated across desktop, web and phone, and built so that reporting a problem never costs a student their safety.

5 / 5
modules integrated
on one cloud database

41
automated tests
run on every push

0
author columns
in the submissions table

Thank you. Questions?

Live: **aura-campus-three.vercel.app**

Code: **github.com/rinshidvp/AURA**